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PRACTICAL PEARL

Bickerstaff's Brainstem Encephalitis Presenting to the ICU

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Abstract

Background Bickerstaff's brainstem encephalitis continues to pose a diagnostic and treatment challenge since the original descriptions by Bickerstaff and Miller-Fisher. The clinical syndrome overlaps with AIDP and MFS, but is accompanied by decreased level of consciousness not attributable to other causes, and the variable presence of long-tract signs.

Methods The methods in this study include the case presentation with autopsy findings of an elderly male presented with progressive weakness and impairment of consciousness and the literature review.

Results Examination revealed areflexia and loss of most brainstem reflexes. Some improvement occurred after several weeks in the ICU, prior to death from pulmonary embolism. Pathologic specimens were similar to others in the literature, with inflammatory changes in nerve roots and brainstem.

Conclusions The above findings led us to conclude that Bickerstaff's brainstem encephalitis remains a clinical diagnosis despite advances in electrophysiologic testing and neuroimaging. BBE likely represents part of a spectrum, overlapping with AIDP and MFS. Immunomodulation may be helpful in shortening the clinical course.

Keywords Bickerstaff · Miller-Fisher · AIDP · Combined demyelination

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Case Summary

An 80-year-old gentleman presented to the emergency department with difficulty with ambulation and generalized weakness progressing over several days. There was equivocal history of chronic lower back pain and possibly some pain in the legs as well. There was no history of preceding febrile illness. There was no history of paresthesias in the extremities. When he became unable to get out of bed independently, he was taken to the emergency room and admitted to the medicine service. While on the ward, he developed features of delirium (disorientation and agitation) and eventually decreased level of consciousness. He required intubation for respiratory failure presumed secondary to decreased consciousness, and the neurocritical care service was consulted on the first day of his ICU stay as he was found to be unrousable after sedation was stopped. Past medical history was significant for coronary artery disease with bypass grafting several years prior, hypertension, hyperlipidemia, remote history of prostate cancer treated with radiation, and smoking.

He showed no response to noxious stimulation administered supraorbitally or to the extremities. Respirations were regular and shallow, and intermittently he spontaneously triggered the ventilator. Vital signs were within normal limits and no vasopressor drugs were required. He showed bifacial weakness and tended to lie with the mouth open, and with the eyes midline and immobile. Pupillary responses were preserved, but the vestibulo-ocular reflex was absent even with cold caloric testing. Corneal reflexes were initially present, but disappeared on subsequent examinations. The jaw jerk was brisk. Cough and gag reflexes were absent. Tone was normal in the right upper extremity and both legs, but the left arm demonstrated persistent rigidity throughout range of motion that was partially velocity-dependent. Reflexes in the extremities were all absent and plantar responses were mute. Heart sounds and there was no evidence of meningism, rash, or adenopathy.

Initial blood work demonstrated no significant abnormalities aside from a mild leukocytosis. While in the ICU the patient developed renal insufficiency, although this was not felt to account for his impairment of consciousness as it was a late finding and responded well to hydration alone. EEG demonstrated mild slowing, and triphasic waves suggestive of metabolic encephalopathy MRI of the brain were essentially normal. MRI of the spine demonstrated possible foci of epidural infection at 2 levels in the thoracic spine, but no evidence of spinal cord compression. Cerebrospinal fluid analysis demonstrated mild lymphocytic pleocytosis (62 wbc/mm^3), and significantly elevated protein (1,513 mg/dl). Glucose was normal. Extensive investigations for infectious agents were negative (HSV 1 and 2 PCR, EBV, CMV, VZV, bacterial cultures, fungal elements, acid-fast bacilli, tuberculosis PCR, and West Nile virus). Cytology and flow cytometry from the CSF were negative for any neoplastic process. Serum and CSF angiotensin converting enzyme levels were normal. Chest X-ray was negative for adenopathy. Nerve conduction studies demonstrated mild slowing and decreased amplitude, with some evidence of denervation on needle electrode examination, suggestive of mixed demyelinating/axonal polyneuropathy. The patient was treated for possible sepsis with broad-spectrum antibiotics. Only mild fevers were documented early in the course of his illness, and blood cultures were negative.

The patient's condition persisted essentially unchanged for almost 7 weeks. During that time, he was treated with intravenous immunoglobulin (IVIG) for 5 days, dexamethasone for 5 days, and received intravenous thiamine. He eventually underwent tracheostomy and gastrostomy tube insertion. He was transferred back to the ward after approximately 6 weeks in the ICU (see video). One week later, eye movements were noted to be improving, and within a few days he was able to speak a few words and was following commands with the right arm. Unfortunately, after approximately 2 months in hospital, the patient unexpectedly died of pulmonary embolism.

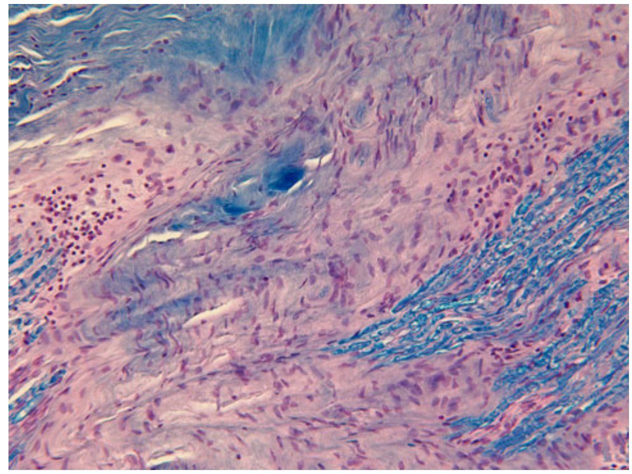

Fig. 1 Trigeminal nerve roots with focal lymphocytic aggregates; LFB/HE ×200

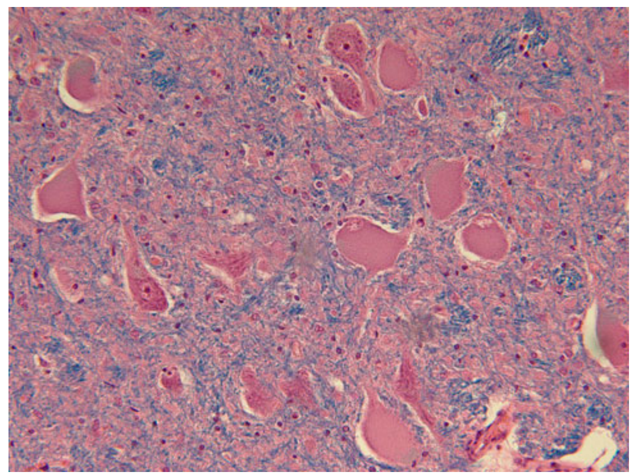

Fig. 2 Central chromatolysis of motor neurons in trigeminal motor nucleus

Pathological Findings (Figs. 1, 2, 3)

Multiple sections through the brainstem showed a mild perivascular lymphocytic infiltrate, consisting of mainly B-lymphocytes (CD20-positive) and scattered microglial nodules. The neurons of oculomotor, trochlear, abducens, facial, and hypoglossal nuclei were significant for central chromatolysis. A mild lymphocytic infiltrate was also identified in the cranial nerves, especially the trigeminal nerve, accompanied by ongoing axonal degeneration and loss. Immunohistochemistry for HSV1, HSV2, EBV, CMV, and varicella was negative in the brainstem sections. Sections from the cervical, lumbar, and sacral spinal cord showed anterior horn neurons with central chromatolysis and a mild lymphocytic infiltrate in the spinal nerve roots and leptomeninges. Few Lewy bodies were noted in the amygdala, nucleus

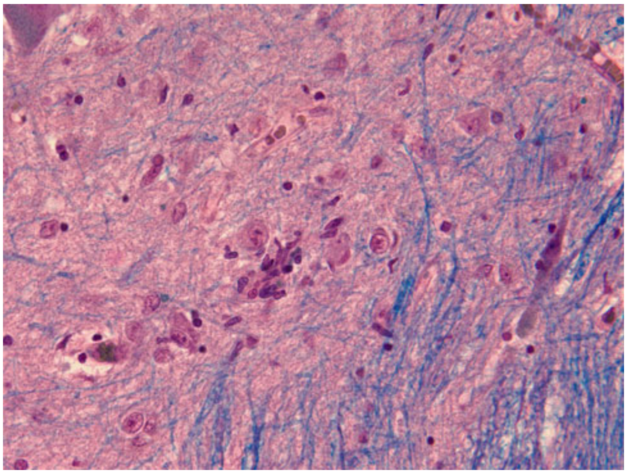

Fig. 3 Migroglial nodule in dorsal motor nucleus of vagus; LFB/HE ×400

basalis, substantia nigra, and locus ceruleus. The pathologic diagnosis was encephalomyeloradiculitis, and taking the clinical setting into consideration, was consistent with Bickerstaff's encephalitis.

Discussion

Between 1951 and 1957, Bickerstaff [1, 2] and Fisher [3] separately described 11 cases of a newly recognized syndrome characterized by variable combinations of ophthalmoplegia, ataxia, impairment of reflexes, and altered consciousness, ranging from drowsiness to coma. Owing to the common findings of prodromal infections and albuminocytologic dissociation, an overlap with Guillain–Barre syndrome has always been suspected, although only one of Bickerstaff's first eight cases had flaccid limb weakness. A summary of the original 11 patients is seen in Table 1. The variability in clinical features made a unified set of diagnostic criteria challenging, and Bickerstaff's opening comments from his 1951 paper highlight the original—and persistent—difficulties in understanding the condition, and the ongoing importance of recognizing clinical syndromes:

> Despite the advances of the recent decades in scientific methods of investigation, conditions are not infrequently seen, particularly in neurological practice, which fall outside the well-recognized syndromes and in which it is possible to hazard only a guess at the etiological pathological processes involved. If the patient recovers, the exact explanation may never be certain, but one experience with such a condition may enable it to be recognized again, and even in the face of ignorance about its nature, some idea of its likely course and prognosis may be formed.

Since then, there have been many reports of overlap syndromes with features of classic Fisher syndrome (FS; ataxia, areflexia, and ophthalmoplegia) and Bickerstaff's brainstem encephalitis (BBE), with the larger series' reported by Al-Din et al. in 1982 (with Bickerstaff as one of the authors) [4], Odaka et al. in 2003 [5], and from Ito et al. in 2008 [6]. In the largest report, the number of patients with BBE was relatively small compared to those with FS (53 vs. 466), but they found that the two groups were relatively similar with respect to (BBE:FS): Serology for anti-GQ1b antibodies (68 vs. 83 %), albuminocytologic dissociation (46 vs. 76 %), MRI abnormalities (11 vs. 2 %), and EEG changes (57 vs. 25 %). One might argue that these differences are more significant than the authors propose, and that these findings are not necessarily "conclusive evidence" of an overlap between FS and BBE as suggested in their discussion. The clinical data are perhaps more convincing, since the median age, infectious prodrome, and initial symptoms were very similar. Reflexes

Table 1 Original 11 cases from Bickerstaff and Fisher (1951–1957) [1–3]

Case	Author, Year	Age, Gender	Prodrome	Consciousness	Ophthalmoplegia	Weakness	Reflexes	CSF
1	B, 1951	24, F	Malaise, headache	Drowsy, rousable	Y	Y (face)	NA	26 wbc
2	B, 1951	36, M	Malaise, headache	Drowsy, rousable	Y	Y (face)	Decreased	ACD
3	B, 1951	24, F	Headache	Drowsy, rousable	Y	Y (face)	NA	N
4	B, 1957	47, F	Vertigo, vomiting, HP	Drowsy	Y	Y (diffuse)	NA	N
5	B, 1957	11, M	Malaise, headache	Drowsy	Y	Y (diffuse)	Absent	12 wbc
6	B, 1957	10, M	Malaise, headache	Drowsy	NA	Y (face)	NA	28 wbc
7	B, 1957	10, M	Malaise, headache	Drowsy	Y	Y (face)	Absent	65 wbc
8	B, 1957	14, M	Malaise, myalgias	Drowsy	Y	Y (face)	Absent	ACD
9	F, 1956		Malaise	NA	Y	Y (diffuse)	Absent	ACD
10	F, 1956		Malaise	Drowsy	Y	NA	Absent	N
11	F, 1956		Headache	NA	Y	NA	Absent	N

B Bickerstaff, *F* Fisher, *ACD* albuminocytologic dissociation, *wbc* white blood cell count, *NA* not assessed

Table 2 Differential diagnosis of Bickerstaff's brainstem encephalitis

Syndrome	Similarities	Differences
Brainstem stroke or hemorrhage	Ophthalmoplegia, coma, UMN signs	Acute onset
		No LMN signs in extremities
		Posturing
		Normal CSF
Wenicke's encephalopathy	Ophthalmoplegia, encephalopathy	Malnutrition, alcohol abuse
	May have impaired reflexes (beriberi)	No UMN signs
	Ataxia	Normal CSF
Botulism	Ophthalmoplegia	Early loss of pupillary responses
	Weakness	Nausea and vomiting
	LMN dysfunction	More than one patient presenting
Myasthenia gravis	Ophthalmoplegia and bulbar symptoms	Normal mental status (unless hypoxic/hypercapnic)
	Weakness	Fluctuating symptoms, asymmetric ophthalmoplegia
		Normal reflexes
		No prodrome
Brainstem neoplasm	Ophthalmoplegia and CN deficits	Usually less acute
	Impaired consciousness	No prodrome
	Gradual onset	Normal CSF
	UMN signs	No LMN signs in extremities/decreased reflexes
ADEM	May involve brainstem	Very abnormal MRI including supratentorial structures
	Prodrome	May have less protein in CSF
	UMN signs	No LMN signs/decreased reflexes
	CSF may be similar	
Meningeal infiltration (sarcoid, neoplastic)	CN palsies	History of cancer, weight loss, systemic symptoms
	Prodrome	Prominent headache
	Impaired consciousness	Less likely signs of neuropathy/decreased reflexes
	Abnormal CSF	Low CSF glucose in neoplastic disease
		Adenopathy on X-ray in sarcoid

ADEM acute disseminated encephalomyelitis, *UMN* upper motor neuron, *LMN* lower motor neuron, *CN* cranial nerve, *CSF* cerebrospinal fluid, *MRI* magnetic resonance imaging

were decreased or absent in all of the FS patients by definition, and 60 % of the BBE group. Extensor plantar responses were found in 38 % of the BBE group and none of the FS patients. Impaired consciousness was appropriately required for a diagnosis of BBE in this study, and 20 % were comatose.

In the largest series of patients with pure BBE, Odaka et al. [5] reported on 62 patients with progressive, relatively symmetric external ophthalmoplegia and ataxia within 4 weeks. Patients were only included if they also demonstrated disturbance of consciousness or hyper-reflexia, and if other causes were excluded. Important alternate diagnostic considerations are shown in Table 2. Altered consciousness was reported in 74 %, with 29 % showing "stupor" or more severe impairment including coma. Reflexes were absent or decreased in 58 %, albuminocytologic dissociation was seen in up to 57 % after the third week. CSF pleocytosis was seen in up to 36 % in the first week, significantly more commonly than in other autoimmune neuropathies such as Guillain–Barre Syndrome (GBS) or MFS. Association with GQ1b antibodies is also supported in this series, with 66 % of patients showing high titers. Electroencephalographic (EEG) studies were abnormal in 70 %, mainly showing diffuse slowing in the theta and delta frequencies. MRI findings were variable with one-third showing non-specific changes in the brainstem, thalamus, cerebellum, and or cerebrum. Neurophysiologic studies strongly supported a primarily axonal process, suggesting some overlap with acute motor axonal neuropathy (AMAN) subtype of GBS.

One patient from this series underwent autopsy, following a clinical course with striking similarities to ours. This was a 37-year-old patient who developed ophthalmoplegia, weakness, areflexia, and coma after an upper-respiratory illness. Coma persisted for 3 weeks. Neurophysiology showed primarily axonal dysfunction, and CSF showed pleocytosis. He was alert by day 33 but suddenly died from respiratory causes on day 45. Histologic findings in the brainstem included perivascular lymphocytic cuffing and glial nodules. Interestingly, this case also demonstrated marked central chromatolysis of the trigeminal motor nucleus cell bodies, and similar findings in anterior horn cells of the spinal cord. Few other autopsy reports are available—Al-Din et al. [4] reported one case showing microglia and lymphocytic cuffing, and Bickerstaff [2] reported one case with only “remarkably scanty” abnormalities.

Pathophysiologic differences between usual “peripheral” autoimmune neuropathies and central manifestations are speculative, but other syndromes may have relevance to understanding BBE. One such condition is severe combined demyelination syndrome, which presents as either typical AIDP followed by symptoms and signs of acute disseminated encephalomyelitis, or vice versa [7]. In some reports, the onset appears to be truly simultaneous [8], whereas in others, including a recent case at our center (unpublished data), there is clearly a delay between the two events. These cases raise the possibility of a “two-hit” pathogenesis, wherein a primary inflammatory process in one compartment could feasibly lead to antigen presentation in the other and a second wave of inflammation against a similar epitope. Given that these cases are rare, studying optimal interventions is difficult, but early aggressive immunosuppression may be warranted. Options include combined IVIG and steroids, or simply a second stage of treatment when a second phase of disease becomes apparent. Interestingly, in most cases of BBE, the cerebral white matter is generally spared [9–12].

Conclusions

Since the original description by two observant clinicians, understanding of the entity known as BBE has changed little—this is perhaps as much a testament to their skill as it is a humbling acknowledgment of our ongoing limitations, despite decades of advances in medical technology and imaging. It seems clear that there is significant overlap between BBE and other autoimmune neuropathies for the reasons cited above, which have now been repeatedly documented. However, the relevant contributions of central and peripheral processes to each of the clinical findings remains unclear, as do the optimal treatments. This case adds significantly to the paucity of available pathologic data in BBE, and the findings are similar to those few cases that have previously reported. We agree that the disorder is probably one form of a spectrum of disorders including all of the acute autoimmune neuropathies, perhaps most closely related to MFS and AMAN. Ultimately, Bickerstaff’s comments remain as true today as for the original cases, and ongoing efforts toward comprehensive understanding are needed.

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